



Measurement and correlation of (liquid + liquid) equilibrium of the azeotrope (cyclohexane + 2-butanone) with different ionic liquids at $T = 298.15$ K

A.B. Pereiro, A. Rodríguez *

Chemical Engineering Department, Vigo University, P.O. Box 36310, Vigo, Spain

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ABSTRACT

Experimental densities, speeds of sound, and refractive indices of the binary mixtures of 2-butanone with cyclohexane and OMIM PF₆ (1-methyl-3-octylimidazolium hexafluorophosphate) were determined from $T = (293.15 \text{ to } 303.15)$ K, since they are necessary to determine the (liquid + liquid) equilibrium. Excess molar volumes, changes of refractive index on mixing, and deviations in isentropic compressibility for the above systems were calculated. Experimental (liquid + liquid) equilibrium of the ternary mixtures {cyclohexane + 2-butanone + 1-hexyl-3-methylimidazolium hexafluorophosphate (HMIM PF₆)} and (cyclohexane + 2-butanone + OMIM PF₆) were carried out to assess the suitability of HMIM PF₆ and OMIM PF₆ as azeotrope breaker of the mixture cyclohexane and 2-butanone. Selectivity and distribution ratio values, derived from the tie lines data, were presented in order to analyze the best separation solvent in a liquid extraction process. Experimental (liquid + liquid) equilibrium data were compared with the correlated values obtained by means of the NRTL and UNIQUAC models.

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1. Introduction

Azeotropic mixtures in industrial processes become a problem because their separation is impossible by ordinary distillation. The (liquid + liquid) separation emerges as a beneficial separation process reducing the energy consumption and the environmental impact. The applications of ionic liquids (ILs) in extraction process have been extensively investigated and reviewed [1–8]. Moreover, ILs have had increasing attention as possible replacement solvents for volatile organic compounds in the development of a sustainable green chemistry.

In this paper, two ILs have been chosen for the separation of the azeotropic mixture (cyclohexane + 2-butanone). This azeotrope is present in the process for purifying grafted polyolefins [9] and its separation is made by membrane in conjunction with a dephlegmation [10]. The selected ILs are formed by the most extensively investigated cation, that is 1-alkyl-3-methylimidazolium, and the hexafluorophosphate (PF₆[−]) as anion. The interest of these ILs have been weakened to avoid the degradation in the presence of water [11] and at high temperatures [12] leading to the formation of HF; consequently, the application of the hexafluorophosphate ILs is restricted to those under water free conditions and moderate temperatures as it is considered in this proposed separation process.

This paper is a continuation of the thermodynamic study of ILs in the physical properties on mixing [13–15] and the determina-

tion of the (liquid + liquid) equilibrium (LLE) [16–19] for designing extractive process involving ILs on an industrial scale for azeotropic mixtures. As the ILs are relatively new, experimental measurements of physical properties and LLE data are scarce but clearly needed.

For the study of (liquid + liquid) separation of azeotropic systems, it is important to know their thermodynamic behaviour. In this paper, densities, refractive indices, and speeds of sound of the binary mixtures of 2-butanone with cyclohexane or OMIM PF₆ from $T = (293.15 \text{ to } 303.15)$ K and atmospheric pressure were measured. Excess and derived properties have been calculated from the experimental data. A function of the mole fraction and temperature polynomial equation [20] was used to fit these quantities. The standard deviations between experimental and calculated values are shown. LLE data of {cyclohexane + 2-butanone + HMIM PF₆ or OMIM PF₆} were determined at $T = 298.15$ K and atmospheric pressure. Likewise, the obtained LLE data were correlated by applying the NRTL [21] and UNIQUAC [22] equations. The selectivity values for the studied ternary systems were determined in order to investigate the ability of these ILs as solvent in liquid extraction processes.

2. Experimental

2.1. Materials

The chemicals were supplied by Fluka for cyclohexane (≥ 0.998 mass fraction purity) and by Merck for 2-butanone (≥ 0.995 mass fraction purity). Chromatographic tests of the organic solvents

* Corresponding author. Tel.: +34 986 812312; fax: +34 986 812380.
E-mail address: a.rodriquez@uvigo.es (A. Rodríguez).

TABLE 1Physical properties of the pure components at $T = 298.15$ K

Component	$\rho/(\text{g} \cdot \text{cm}^{-3})$		n_D	
	This work	Literature	This work	Literature
Cyclohexane	0.7738	0.7739 [24]	1.42354	1.4235 [24]
2-Butanone	0.7997	0.79974 [25]	1.37618	1.37685 [26]
HMIM PF ₆	1.2937	1.2935 [27]	1.41787	
OMIM PF ₆	1.2357	1.23684 [28]	1.42302	1.423 [29]

showed purities that fulfilled purchaser specifications. Before using, the solvents were degassed ultrasonically, dried over freshly activated molecular sieves (type $4 \cdot 10^{-8}$ m, supplied by Aldrich) for several weeks, and kept under an inert argon atmosphere as soon as the bottles were opened. The ILs were synthesized in our laboratory [23] and characterized by its NMR spectra and positive FABMS (FISONS VG AUTOSPEC mass spectrometer). To reduce the water content to negligible values (mass fraction lower than 0.02%, determined using a 756 Karl Fisher coulometer), vacuum $P = (0.2 \text{ Pa})$ was applied to the ILs during several days, always immediately prior to its use. The ILs were kept in bottles under an inert gas. The physical properties of the pure components at

$T = 298.15$ K are listed in table 1, together with recent literature values [24–29].

2.2. Apparatus and procedure

The mixtures were prepared by transferring via syringe known masses of the pure liquids into stoppered bottles to prevent evaporation, using a Mettler AX-205 Delta Range balance with a precision of $\pm 10^{-5}$ g. The estimated uncertainty on the composition measurement was $\pm 10^{-4}$ mole fraction. Vials are closed with screw caps to ensure a secure seal. The sample is taken from the vial with a syringe through a silicone septum and is immediately put into the apparatus.

2.2.1. Densities and speeds of sound

The density and speed of sound of the reagents and the mixtures were measured with an Anton Paar DSA-48 digital vibrating tube densimeter. The uncertainty in the experimental measurements has been found to be lower than $\pm 2 \cdot 10^{-4} \text{ g} \cdot \text{cm}^{-3}$ for the density and $\pm 1 \text{ m} \cdot \text{s}^{-1}$ for the speed of sound. The apparatus was calibrated by measuring the density of Millipore quality water and ambient air according to the instructions.

TABLE 2Density, ρ , excess molar volume, V_m^E , refractive index, n_D , changes of refractive index on mixing, Δn_D , speed of sound, u , isentropic compressibility, κ_s , and deviation in isentropic compressibility, $\Delta \kappa_s$ at several temperatures

x	$\rho/(\text{g} \cdot \text{cm}^{-3})$	$V_m^E/(\text{cm}^3 \cdot \text{mol}^{-1})$	n_D	Δn_D	$u/(\text{m} \cdot \text{s}^{-1})$	$\kappa_s/(\text{T} \cdot \text{Pa}^{-1})$	$\Delta \kappa_s/(\text{T} \cdot \text{Pa}^{-1})$
$\{x \text{ cyclohexane} + (1-x) \text{ 2-butanone}\}$							
$T = 293.15 \text{ K}$							
0	0.8056	0	1.37879	0	1212	845	0
0.0472	0.8026	0.163	1.38079	−0.0002	1212	848	6
0.0979	0.7998	0.304	1.38290	−0.0005	1213	850	11
0.1986	0.7947	0.540	1.38711	−0.0011	1212	857	24
0.2973	0.7903	0.728	1.39144	−0.0015	1212	861	34
0.3975	0.7866	0.853	1.39592	−0.0018	1216	860	39
0.4970	0.7837	0.901	1.40055	−0.0019	1221	856	41
0.6045	0.7810	0.892	1.40575	−0.0018	1229	848	40
0.6922	0.7795	0.819	1.41006	−0.0016	1236	840	36
0.7984	0.7782	0.654	1.41530	−0.0014	1248	825	28
0.8978	0.7778	0.398	1.42041	−0.0011	1261	809	18
0.9476	0.7780	0.220	1.42316	−0.0007	1269	798	10
1	0.7784	0	1.42633	0	1279	785	0
$T = 298.15 \text{ K}$							
0	0.7997	0	1.37618	0	1192	880	0
0.0472	0.7964	0.209	1.37811	−0.0003	1190	887	10
0.0979	0.7937	0.351	1.38016	−0.0006	1190	890	17
0.1986	0.7888	0.587	1.38436	−0.0012	1190	896	28
0.2973	0.7848	0.752	1.38857	−0.0016	1190	900	39
0.3975	0.7814	0.866	1.39290	−0.0019	1192	901	45
0.4970	0.7784	0.916	1.39774	−0.0020	1198	895	45
0.6045	0.7761	0.913	1.40260	−0.0019	1204	889	45
0.6922	0.7746	0.825	1.40723	−0.0017	1212	879	41
0.7984	0.7734	0.665	1.41250	−0.0015	1222	866	34
0.8978	0.7731	0.405	1.41769	−0.0011	1236	847	22
0.9476	0.7732	0.236	1.42035	−0.0007	1244	836	14
1	0.7738	0	1.42354	0	1256	819	0
$T = 303.15 \text{ K}$							
0	0.7949	0	1.37355	0	1163	931	0
0.0472	0.7915	0.221	1.37540	−0.0004	1159	941	12
0.0979	0.7888	0.361	1.37748	−0.0007	1159	944	19
0.1986	0.7840	0.593	1.38159	−0.0013	1158	951	31
0.2973	0.7799	0.771	1.38588	−0.0017	1158	956	42
0.3975	0.7764	0.893	1.39029	−0.0020	1161	956	47
0.4970	0.7735	0.947	1.39494	−0.0021	1166	952	49
0.6045	0.7711	0.928	1.40014	−0.0019	1172	944	47
0.6922	0.7696	0.856	1.40443	−0.0018	1179	935	43
0.7984	0.7685	0.685	1.40970	−0.0015	1188	921	35
0.8978	0.7682	0.418	1.41473	−0.0012	1200	904	23
0.9476	0.7685	0.237	1.41746	−0.0008	1207	893	15
1	0.7690	0	1.42073	0	1219	875	0

(continued on next page)

Table 2 (continued)

x	$\rho/(\text{g} \cdot \text{cm}^{-3})$	$V_m^E/(\text{cm}^3 \cdot \text{mol}^{-1})$	n_D	Δn_D	$u/(\text{m} \cdot \text{s}^{-1})$	$\kappa_s/(\text{T} \cdot \text{Pa}^{-1})$	$\Delta\kappa_s/(\text{T} \cdot \text{Pa}^{-1})$
$\{x \text{ OMIM PF}_6 + (1-x) \text{ 2-butanone}\}$ $T = 293.15 \text{ K}$							
0	0.8056	0	1.37879	0	1212	845	0
0.0475	0.8698	−0.748	1.38881	0.0079	1243	744	−80
0.0972	0.9237	−1.204	1.39567	0.0124	1268	674	−128
0.1918	1.0020	−1.689	1.40468	0.0171	1303	588	−171
0.2947	1.0631	−1.870	1.41058	0.0184	1331	531	−182
0.3949	1.1079	−1.897	1.41468	0.0179	1352	494	−175
0.4983	1.1436	−1.786	1.41774	0.0162	1368	467	−155
0.5752	1.1649	−1.606	1.41934	0.0143	1378	452	−136
0.6815	1.1895	−1.318	1.42126	0.0114	1389	436	−105
0.7683	1.2058	−0.983	1.42239	0.0086	1399	424	−78
0.8315	1.2164	−0.749	1.42355	0.0068	1406	416	−57
0.9315	1.2309	−0.326	1.42474	0.0035	1425	400	−29
1	1.2396	0	1.42440	0	1424	398	0
$T = 298.15 \text{ K}$							
0	0.7997	0	1.37618	0	1192	880	0
0.0475	0.8643	−0.800	1.38663	0.0082	1224	773	−85
0.0972	0.9189	−1.329	1.39445	0.0137	1249	698	−136
0.1918	0.9976	−1.848	1.40289	0.0177	1286	606	−183
0.2947	1.0588	−1.999	1.40877	0.0188	1315	546	−195
0.3949	1.1037	−2.015	1.41302	0.0183	1337	507	−187
0.4983	1.1394	−1.890	1.41614	0.0166	1354	479	−166
0.5752	1.1609	−1.710	1.41779	0.0147	1364	463	−146
0.6815	1.1855	−1.394	1.41979	0.0117	1376	446	−113
0.7683	1.2020	−1.060	1.42095	0.0088	1386	433	−84
0.8315	1.2128	−0.850	1.42198	0.0069	1392	425	−62
0.9315	1.2275	−0.442	1.42337	0.0036	1410	410	−31
1	1.2357	0	1.42302	0	1408	408	0
$T = 303.15 \text{ K}$							
0	0.7949	0	1.37355	0	1163	931	0
0.0475	0.8596	−0.824	1.38505	0.0092	1195	814	−92
0.0972	0.9141	−1.340	1.39284	0.0146	1229	724	−157
0.1918	0.9932	−1.909	1.40124	0.0185	1269	625	−207
0.2947	1.0546	−2.071	1.40709	0.0194	1300	561	−218
0.3949	1.0995	−2.072	1.41133	0.0188	1323	520	−208
0.4983	1.1353	−1.935	1.41466	0.0171	1340	490	−185
0.5752	1.1569	−1.753	1.41645	0.0152	1351	474	−162
0.6815	1.1815	−1.430	1.41825	0.0119	1363	456	−126
0.7683	1.1983	−1.126	1.41949	0.0090	1372	443	−94
0.8315	1.2091	−0.907	1.42051	0.0070	1379	435	−70
0.9315	1.2239	−0.494	1.42199	0.0036	1395	420	−34
1	1.2319	0	1.42165	0	1393	418	0

TABLE 3

Fitting parameters A_{ij} and standard deviations (σ) calculated from equations (4) and (5), respectively

$\{x \text{ cyclohexane} + (1-x) \text{ 2-butanone}\}$						
$V_m^E/(\text{cm}^3 \cdot \text{mol}^{-1})$	$A_{11} = 3.626$ $A_{12} = 0.079$ $A_{13} = 0.095$	$A_{21} = 0.461$ $A_{22} = -0.056$ $A_{23} = 0.103$	$A_{31} = 0.201$ $A_{32} = 0.046$ $A_{33} = -0.165$	$A_{41} = 0.142$ $A_{42} = -1.297$ $A_{43} = 0.767$	$A_{51} = 0.332$ $A_{52} = 2.170$ $A_{53} = -1.230$	$\sigma = 0.005$
Δn_D	$A_{11} = -0.0074$ $A_{12} = -0.0013$ $A_{13} = 0.0005$	$A_{21} = 0.0001$ $A_{22} = 0.0003$ $A_{23} = 0.0009$	$A_{31} = -0.0003$ $A_{32} = 0.0005$ $A_{33} = 0.0004$	$A_{41} = -0.0046$ $A_{42} = 0.0011$ $A_{43} = -0.0048$	$A_{51} = -0.0033$ $A_{52} = -0.0004$ $A_{53} = -0.0028$	$\sigma = 0.00003$
$\Delta\kappa_s/(\text{T} \cdot \text{Pa}^{-1})$	$A_{11} = 162$ $A_{12} = 58$ $A_{13} = -26$	$A_{21} = 0.5$ $A_{22} = 55$ $A_{23} = -58$	$A_{31} = 9.0$ $A_{32} = 3.4$ $A_{33} = -23$	$A_{41} = 66$ $A_{42} = -90$ $A_{43} = 77$	$A_{51} = -14$ $A_{52} = 94$ $A_{53} = 37$	$\sigma = 0.6$
$\{x \text{ OMIM PF}_6 + (1-x) \text{ 2-butanone}\}$						
$V_m^E/(\text{cm}^3 \cdot \text{mol}^{-1})$	$A_{11} = -7.119$ $A_{12} = -0.742$ $A_{13} = 0.144$	$A_{21} = 3.23$ $A_{22} = 0.922$ $A_{23} = -0.114$	$A_{31} = -1.279$ $A_{32} = 0.353$ $A_{33} = -1.378$	$A_{41} = 3.326$ $A_{42} = -2.293$ $A_{43} = 0.617$	$A_{51} = -3.741$ $A_{52} = -5.020$ $A_{53} = 3.862$	$\sigma = 0.009$
Δn_D	$A_{11} = 0.0648$ $A_{12} = 0.0026$ $A_{13} = 0.0012$	$A_{21} = -0.0415$ $A_{22} = 0.0011$ $A_{23} = -6.4\text{E-}05$	$A_{31} = 0.0201$ $A_{32} = 0.0009$ $A_{33} = -0.0080$	$A_{41} = -0.0242$ $A_{42} = -0.0188$ $A_{43} = -0.0019$	$A_{51} = 0.0475$ $A_{52} = 0.0171$ $A_{53} = 0.0137$	$\sigma = 0.00009$
$\Delta\kappa_s/(\text{T} \cdot \text{Pa}^{-1})$	$A_{11} = -621$ $A_{12} = -55$ $A_{13} = -64$	$A_{21} = 444$ $A_{22} = 18$ $A_{23} = 69$	$A_{31} = -207$ $A_{32} = 1.1$ $A_{33} = -116$	$A_{41} = 308$ $A_{42} = -20$ $A_{43} = 97$	$A_{51} = -457$ $A_{52} = 5.3$ $A_{53} = 1.2$	$\sigma = 0.8$

2.2.2. Refractive indices

The refractive indices were determined by the automatic refractometer ABBEMAT-WR Dr. Kernchen with an uncertainty in the experimental measurements of $\pm 4 \cdot 10^{-5}$. The apparatus was calibrated by measuring the refractive index of Millipore quality water and tetrachloroethylene (supplied by the company) before each series of measurements according to instructions.

2.2.3. Determination of LLE

Density and refractive index calibration curves of the coexisting phase were made with known composition samples of the ternary mixture to determine the mole fraction at $T = 298.15$ K. The uncertainty of the phase composition is estimated to be ± 0.005 in mole fraction. This method was described in the literature [18,19]. Tie-line data determination was performed in a jacketed glass vessel containing a magnetic stirrer connected to a temperature controlled circulating bath (controlled to ± 0.01 K). The vessel was closed to moisture and was flushed with dry argon. The temperature in the cell was measured with a F200 ASL digital thermometer with an uncertainty of ± 0.01 K. The measurements of the ternary LLE started with the addition of 30 cm³ of immiscibility ternary components of known composition to the vessel, the temperature was adjusted and the mixture was stirred vigorously during 1 h and left to settle for 4 h. Samples were taken by a syringe from

the upper and lower phases. A series of LLE measurements were made over the entire composition range.

The LLE was determined by two techniques: the experimental determination of the tie-lines that shows the composition of each layer and the determination of the binodal curve for the immiscible area of the ternary system.

3. Results and discussion

The density, refractive index, speed of sound, excess molar volume, changes of refractive index on mixing, isentropic compressibility, determined by means of Laplace equation, ($\kappa_S = \rho^{-1} \cdot u^{-2}$) and deviation in isentropic compressibility of (cyclohexane + 2-butanone) and (OMIM PF₆ + 2-butanone) from $T = (293.15$ to $303.15)$ K are given in table 2. The physical properties of (HMIM PF₆ + 2-butanone) were published previously [13]. The binary mixture (cyclohexane + 2-butanone) presents azeotropic behaviour at $P = 101.3$ kPa and the azeotropic composition [30] is $x_1 = 0.52$.

Excess molar volumes, changes of refractive index on mixing, and deviations in isentropic compressibility were calculated from the experimental values, as follows:

$$V_m^E = \sum_{i=1}^N x_i M_i (\rho^{-1} - \rho_i^{-1}), \quad (1)$$

$$\Delta n_D = n_D - \sum_{i=1}^N x_i n_{Di}^0, \quad (2)$$

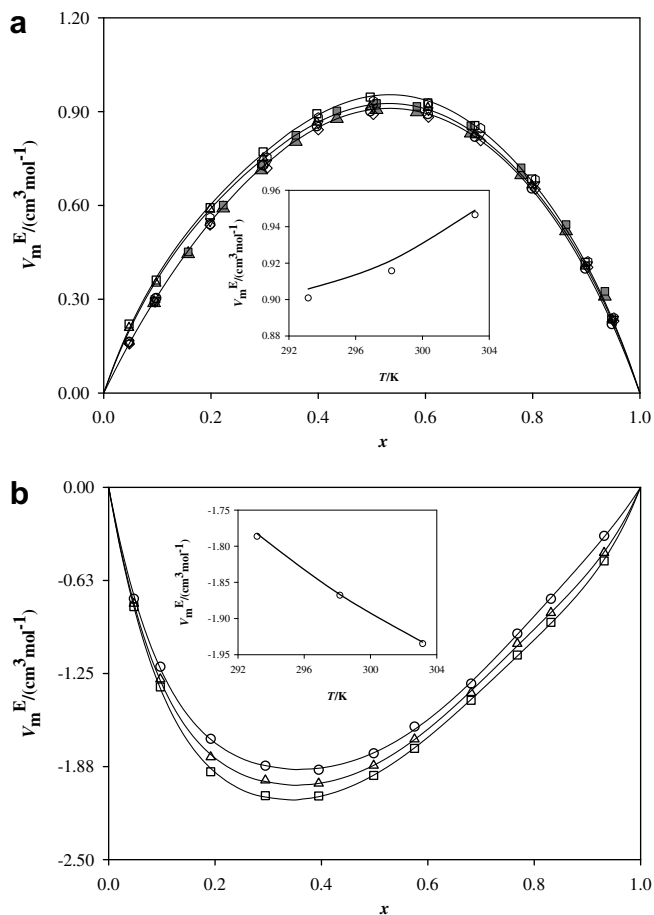


FIGURE 1. Plot of excess molar volume, V_m^E , versus mole fraction, x , for the binary mixtures: (a) { x cyclohexane + $(1-x)$ 2-butanone}: $\circ$, this work and $\diamond$, Gonzalez *et al.* [31] at $T = 293.15$ K; $\triangle$, this work, ∇ , Gonzalez *et al.* [31] and, $\blacktriangle$, Lee *et al.* [32] at $T = 298.15$ K; $\square$, this work, $\circ$ Gonzalez *et al.* [31] and, $\blacksquare$, Lee *et al.* [32] at $T = 303.15$ K. (b) { x OMIM PF₆ + $(1-x)$ cyclohexane}: $\circ$, this work at $T = 293.15$ K; $\triangle$, this work at $T = 298.15$ K; $\square$, this work at $T = 303.15$ K and fitted curve. Inside figures excess molar volumes of the binary mixtures against the temperature at equimolar composition.

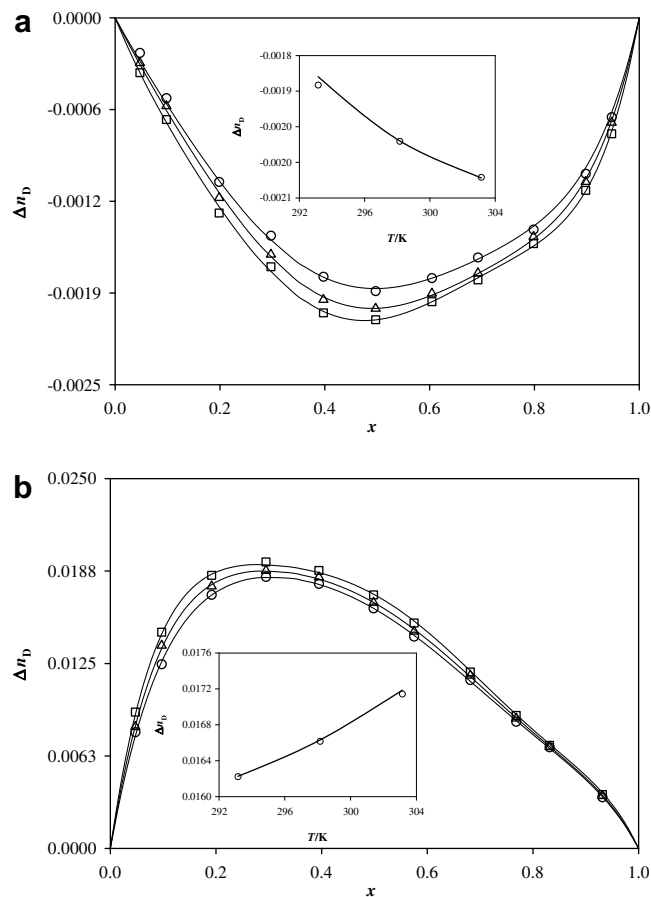


FIGURE 2. Plot of changes of refractive indices on mixing Δn_D of the binary mixtures against mole fraction x for: (a) { x cyclohexane + $(1-x)$ 2-butanone} and (b) { x OMIM PF₆ + $(1-x)$ cyclohexane} $\circ$, this work at $T = 293.15$ K; $\triangle$, this work at $T = 298.15$ K; $\square$, this work at $T = 303.15$ K and fitted curve. Inside figures excess molar volumes of the binary mixtures against the temperature at equimolar composition.

$$\Delta\kappa_S = \kappa_S - \sum_{i=1}^N x_i \kappa_{S,i}. \quad (3)$$

In these equations, ρ and n_D are the density and refractive index of the mixture, respectively, ρ_i^o and $n_{D,i}^o$ are the density and refractive index of the pure components, respectively, κ_S is the isentropic compressibility of the mixture and $\kappa_{S,i}$ is the isentropic compressibility of the pure components.

The excess and derived quantities of the binary mixtures were fitted to a function of the mole fraction and temperature polynomial equation developed as follows:

$$\Delta Q = x \cdot (1-x) \cdot \sum_{i=1}^5 \sum_{j=1}^3 A_{ij} \cdot 10^{1-j} (2x-1)^{i-1} \cdot (T-293.15)^{j-1}, \quad (4)$$

where ΔQ is the excess or derived property, x is the mole fraction and A_{ij} is the fitting parameter. Applying the SOLVER function in Microsoft EXCEL, the degree of polynomial expression was optimized. The correlation parameters are listed in table 3, together with the standard deviations. These deviations were calculated by applying the following expression:

$$\sigma = \left(\sum_i^{n_{\text{DAT}}} (z_{\text{exp}} - z_{\text{pred}})^2 / n_{\text{DAT}} \right)^{1/2}, \quad (5)$$

TABLE 4

Composition of the experimental tie-lines, solute distribution ratio, β , and selectivity, S , for the ternary systems at $T = 298.15$ K

Organic-rich phase		Ionic liquid-rich phase		β	S
x_1^I	x_2^I	x_1^{II}	x_2^{II}		
{Cyclohexane (1) + 2-butanone (2) + HMIM PF ₆ (3)}					
0.985	0.015	0.071	0.128	8.31	114.85
0.961	0.039	0.079	0.217	5.56	67.99
0.947	0.053	0.083	0.293	5.53	63.40
0.919	0.081	0.093	0.366	4.51	44.52
0.867	0.132	0.110	0.416	3.14	24.68
0.815	0.185	0.135	0.476	2.58	15.52
0.788	0.211	0.154	0.514	2.43	12.44
0.750	0.249	0.180	0.558	2.24	9.33
0.695	0.304	0.199	0.587	1.93	6.74
0.660	0.338	0.215	0.600	1.77	5.45
0.629	0.369	0.237	0.606	1.64	4.36
{Cyclohexane (1) + 2-butanone (2) + OMIM PF ₆ (3)}					
0.991	0.009	0.209	0.108	11.66	55.30
0.974	0.026	0.209	0.177	6.77	31.56
0.959	0.041	0.210	0.240	5.90	26.93
0.942	0.058	0.215	0.303	5.23	22.88
0.922	0.078	0.221	0.362	4.62	19.27
0.897	0.103	0.236	0.409	3.98	15.13
0.852	0.148	0.261	0.461	3.12	10.19
0.778	0.221	0.296	0.503	2.27	5.97
0.708	0.290	0.328	0.530	1.83	3.94
0.640	0.356	0.357	0.538	1.51	2.71

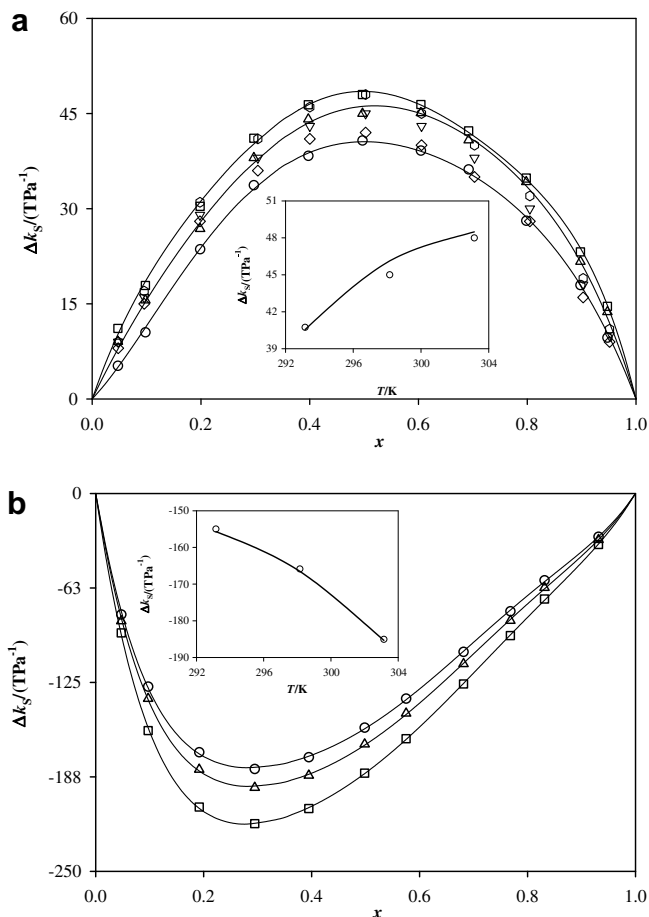


FIGURE 3. Plot of the deviations in isentropic compressibility $\Delta\kappa_S$ of the binary mixtures against mole fraction x for: (a) {x cyclohexane + (1-x) 2-butanone}, $\circ$, this work and $\diamond$, Gonzalez *et al.* [31] at $T = 293.15$ K; $\triangle$, this work and ∇ , Gonzalez *et al.* [31] at $T = 298.15$ K; $\square$, this work and F[x], Gonzalez *et al.* [31] at $T = 303.15$ K. (b) {x OMIM PF₆ + (1-x) cyclohexane}, $\circ$, this work at $T = 293.15$ K; $\triangle$, this work at $T = 298.15$ K; $\square$, this work at $T = 303.15$ K and fitted curve. Inside figures excess molar volumes of the binary mixtures against the temperature at equimolar composition.

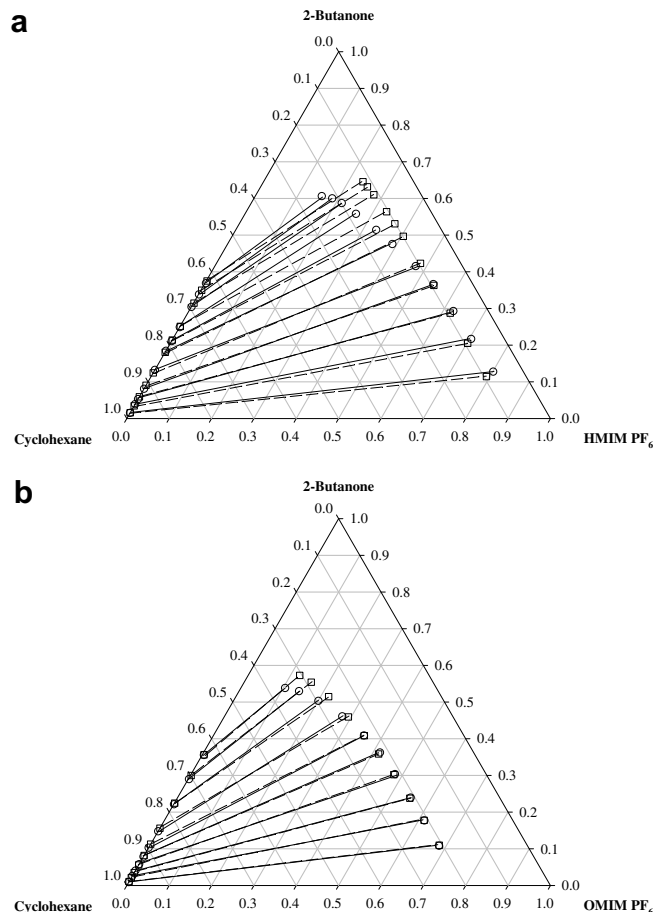


FIGURE 4. Experimental tie-lines of the ternary systems at $T = 298.15$ K: (a) {cyclohexane (1) + 2-butanone (2) + HMIM PF₆ (3)}: $\circ$ and solid line, this work, $\square$ and dashed line, UNIQUAC correlation. (b) {cyclohexane (1) + 2-butanone (2) + OMIM PF₆ (3)}: $\circ$ and solid line, this work, $\square$ and dashed line, NRTL correlation.

where property values and the number of experimental data are represented by z and n_{DAT} , respectively.

Excess molar volumes for the binary mixtures (cyclohexane + 2-butanone) and (OMIM PF₆ + 2-butanone) and the fitted curves from $T = (293.15 \text{ to } 303.15) \text{ K}$ versus the mole fraction are shown in figure 1. The comparison with literature [31,32] data is enclosed and the results agree with the experimental data. Excess molar volumes are positive over the entire composition range for (cyclohexane + 2-butanone). However, opposite behaviour is observed for the binary mixture (OMIM PF₆ + 2-butanone).

On the other hand, changes of refractive index on mixing for the binary mixtures plotted against mole fraction together with the fitted curves at the studied temperatures are shown in figure 2. These quantities are negatives for (cyclohexane + 2-butanone) and positives for (OMIM PF₆ + 2-butanone) over the whole composition range. The deviations in isentropic compressibility show the same behaviour than excess molar volumes in figure 3 over the entire composition range. The comparison with the literature [31] data is enclosed.

A study of the temperature effect over the derived properties at equimolar composition of the binary mixtures is made and the results are shown in figures 1–3. When the temperature is increased for (cyclohexane + 2-butanone), the excess molar volume and deviations in isentropic compressibility increase, while the refractive

index on mixing decreases. However, when we increase it for (OMIM PF₆ + 2-butanone), the behaviour is opposite, the excess molar volume and deviations in isentropic compressibility decrease, whereas the changes in refractive index on mixing increase.

The composition of the experimental tie-line ends for the ternary mixtures of (cyclohexane + 2-butanone + HMIM PF₆ or OMIM PF₆) are performed at $T = 298.15 \text{ K}$ and atmospheric pressure in table 4. The corresponding triangular diagrams with the experimental tie-lines for the systems are shown in figure 4. An examination of this figure indicates a clear idea of the shape and the size of the immiscibility region of the systems. The positive slope shows that the solute goes preferentially to the solvent-rich phase. Another significant aspect is the fact the IL does not enter in the equilibrium organic-rich phase, that is to say, the presence of HMIM PF₆ or OMIM PF₆ was not detected in the upper phase.

3.1. Solute distribution and selectivity

Values of solute distribution ratio, β , and selectivity, S , are shown in table 4. These parameters are defined as follows:

$$\beta = x_2^{\text{II}}/x_2^{\text{I}}, \quad (6)$$

$$S = (x_1^{\text{I}}/x_1^{\text{II}}) \cdot (x_2^{\text{II}}/x_2^{\text{I}}). \quad (7)$$

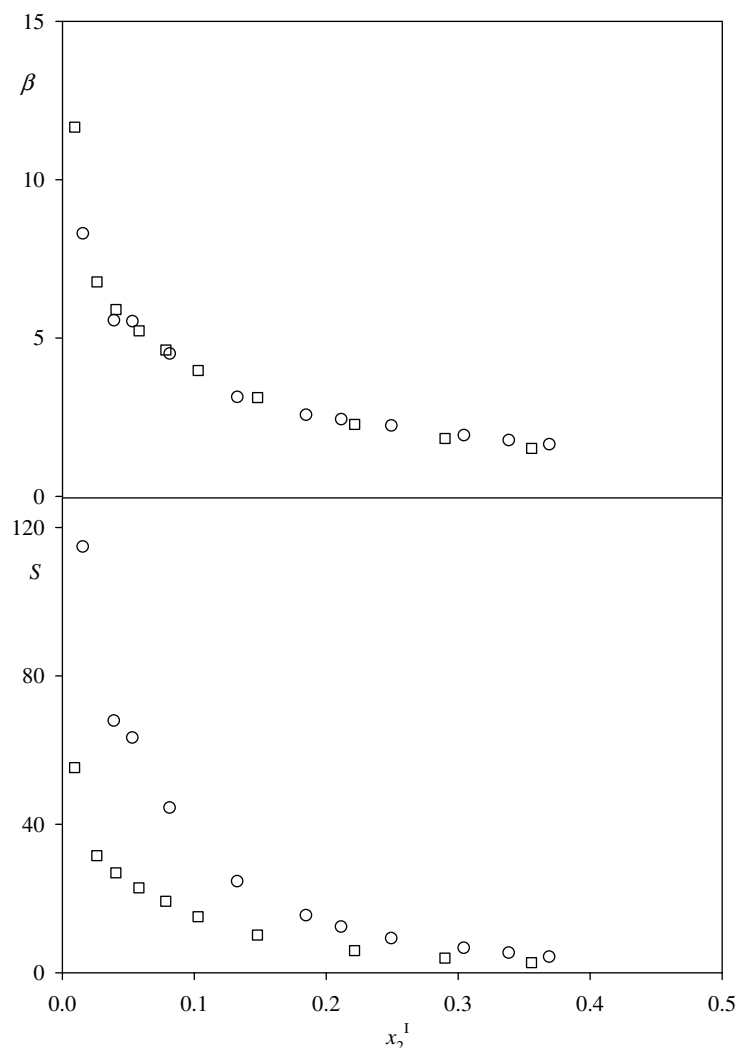


FIGURE 5. Plot of solute distribution ratio, β , and selectivity, S , against mole fraction of the systems presenting azeotrope (cyclohexane + 2-butanone) with: ○, HMIM PF₆ and □, OMIM PF₆ versus 2-butanone mole fraction in the organic phases at $T = 298.15 \text{ K}$.

In these equations, x is the mole fraction, subscripts 1 and 2 indicate the cyclohexane and the 2-butanone, respectively, and superscripts I and II indicate the organic and IL phases, respectively.

The values of the solute distribution ratio and selectivity for the studied ternary systems as a function of the solute concentration in the organic phase are plotted in figure 5. The solute distribution ratio provides the solvent capacity of the IL, related to the amount of IL required for the process. The selectivity is an important parameter to assess the efficiency of the IL used as solvent in the selective extraction of the solute from the azeotropic system. In this case, the selectivity values for all areas in the binodal curves are higher than the unit, from which it can be inferred that the extraction of the solute from the azeotropic system is indeed possible. A comparison between the selectivity values for the azeotropic system shows that the HMIM PF₆ obtains higher values than OMIM PF₆ for the proposal process.

3.2. Correlation of LLE

In this work, the NRTL and UNIQUAC equations are used to correlate the experimental LLE data. For the NRTL model, the third randomness parameter α was optimized and their values are reported in table 5. The volume R_k and surface area Q_k parameters of the UNIQUAC equation were obtained from the literature [33]. In these equations, the adjustable parameter was defined as follows for NRTL and UNIQUAC, respectively:

$$\tau_{ji} = \Delta g_{ij}/R \cdot T, \quad (8)$$

$$\tau_{ji} = \exp(-\Delta u_{ij}/R \cdot T). \quad (9)$$

The DISTILL 3.0 spreadsheet [34] was employed to minimize the difference between the experimental and calculated mole fraction defined as:

$$\text{OF} = \sum_{i=1}^n [(x_{1i}^I - x_{1i}^I(\text{calc}))^2 + (x_{2i}^I - x_{2i}^I(\text{calc}))^2 + (x_{1i}^{II} - x_{1i}^{II}(\text{calc}))^2 + (x_{2i}^{II} - x_{2i}^{II}(\text{calc}))^2], \quad (10)$$

where x_{1i}^I , x_{2i}^I , x_{1i}^{II} , x_{2i}^{II} are the experimental mole fraction; $x_{1i}^I(\text{calc})$, $x_{2i}^I(\text{calc})$, $x_{1i}^{II}(\text{calc})$ and $x_{2i}^{II}(\text{calc})$ are the calculated mole

TABLE 5

Fitting parameters and standard deviation of the LLE data by means of the NRTL equation

System	ij	$\Delta g_{ij}/(\text{J} \cdot \text{mol}^{-1})$	$\Delta g_{ji}/(\text{J} \cdot \text{mol}^{-1})$	α	σ
{Cyclohexane (1) + 2-butanone (2) + HMIM PF ₆ (3)}	12	418,680	355.33	0.3	0.042
	13	5977.5	20,899		
	23	-8518.9	13019		
{Cyclohexane (1) + 2-butanone (2) + OMIM PF ₆ (3)}	12	1984773	-401.12	0.1	0.012
	13	-5485.5	33,025		
	23	-13,664	9959.6		

TABLE 6

Fitting parameters and standard deviation of the LLE data by means of the UNIQUAC equation

System	ij	$\Delta u_{ij}/(\text{J} \cdot \text{mol}^{-1})$	$\Delta u_{ji}/(\text{J} \cdot \text{mol}^{-1})$	σ
{Cyclohexane (1) + 2-butanone (2) + HMIM PF ₆ (3)}	12	3479.2	-4038.1	0.032
	13	333.27	2612.8	
	23	-3869.9	-245.56	
{Cyclohexane (1) + 2-butanone (2) + OMIM PF ₆ (3)}	12	99721	-89091	0.015
	13	-778.33	3848.3	
	23	-88,982	25.461	

TABLE 7

Plait point values from the best correlation equation for the ternary systems at $T = 298.15 \text{ K}$

x_1	x_2	x_3
{Cyclohexane (1) + 2-butanone (2) + HMIM PF ₆ (3)} (UNIQUAC)		
0.257	0.698	0.045
{Cyclohexane (1) + 2-butanone (2) + OMIM PF ₆ (3)} (NRTL)		
0.424	0.544	0.032

fraction; and superscripts I and II indicate the organic and IL phases, respectively.

The fitting parameters of the equation are presented in tables 5 and 6 together with the standard deviations. These deviations were calculated by applying the following expression:

$$\sigma = \left(\sum_i (x_{ilm}^{\text{exp}} - x_{ilm}^{\text{calc}})^2 / 6k \right)^{1/2}, \quad (11)$$

where x is the mole fraction and the subscripts i , l and m provide the component, the phase and the tie-line, respectively. The k value refers to the number of interaction components.

A view of the standard deviation between the experimental data and those obtained from their correlation by the NRTL and UNIQUAC equations indicates that last one is more adequately to correlate the immiscible region of these mixtures. A comparison between the experimental data and those obtained from the best correlated model for the ternary mixtures (cyclohexane + 2-butanone + HMIM PF₆ or OMIM PF₆) are shown in figure 4. The better correlation results were obtained for OMIM PF₆ than for HMIM PF₆, and the organic phase is fitted better than IL phase. Plait point values have been obtained from the best correlated model and are presented in table 7.

4. Conclusions

Cyclohexane and 2-butanone form an azeotrope due to the non-polar character of cyclohexane with respect to the slight polar nature of 2-butanone and, the cyclic configuration of the first compound compared to the aliphatic and asymmetric structure of the second one. All these differences result in a kind of repulsive interaction between the two compounds so that the miscibility of both was mainly assumed by the entropic contribution, which increases with increasing temperature. This research has been focused on a comparative study of two imidazolium-based ILs with hexafluorophosphate anion as solvents in (liquid + liquid) extraction. Density, refractive index and speed of sound as a function of the temperature for the binary mixtures of 2-butanone with cyclohexane and OMIM PF₆ are presented, as they are necessary to determine the (liquid + liquid) equilibrium. The positive excess molar volume values indicate that the interactions between the cyclohexane with 2-butanone are smaller than those between molecules of each pure component. However, the binary mixture (OMIM PF₆ + 2-butanone) exhibits negative values due to the interactions between the IL and 2-butanone are stronger than those between cyclohexane and 2-butanone ones. The weak electrolytic character of the two ILs compounds is very compatible with the polar nature of 2-butanone; this is fully supported by the largely negative values of the excess molar volumes for the mixture (OMIM PF₆ + 2-butanone). Also, this supported the higher distribution coefficient of 2-butanone in the IL phase.

LLE data were obtained for the mixtures of (cyclohexane + 2-butanone + HMIM PF₆) and (cyclohexane + 2-butanone + OMIM PF₆) at $T = 298.15 \text{ K}$ and atmospheric pressure. Likewise, the experimental data have been successfully correlated by applying the NRTL and UNIQUAC equations, thus facilitating their implementa-

tion and use in computerized applications for the design of a separation process. The results show that the best correlation equation is the UNIQUAC model, because of this equation obtains better results for the solutions with relatively strong intermolecular interactions. The use of HMIM PF₆ as solvent leads to higher values of selectivity than the OMIM PF₆, indicating that it would be a better choice as solvent in this separation process. The analysis of the data indicate that the alkyl chain length of the imidazolium ring plays a negative role in the capability of the RMIM PF₆ to purify the cyclohexane by a liquid extractive process.

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